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Simulation & Modelling Course Project: 3D Mass-Spring-Damper Cloth Simulation

by

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Abstract: This report presents a real-time 3D cloth simulation using a mass-spring-damper model implemented in Python with VPython for visualization. The cloth is represented as a grid of particles connected by structural, shear, and bend springs following Provot's topology, with forces computed from Hooke's Law. Forward Euler and fourth-order Runge-Kutta (RK4) are implemented and compared. Stability testing on a coupled system of over 600 springs with natural frequency $\omega \approx 63$ rad/s demonstrates that Euler's method diverges at time-steps as small as $dt = 0.005$, below the theoretical single-spring limit of $dt < 0.032$, due to compounding error across coupled particles. RK4 remains stable up to $dt = 0.01$, providing a much wider time-step range. The simulation supports real-time interactive control of gravity, wind, particle mass, and spring constants, and a vectorized force computation enables 60fps performance on grids of over 200 particles.

1 Introduction

Cloth simulation using physically based modeling dates back to Terzopoulos et al. [4], with the mass-spring-damper approach formalized by Provot [3]. Since its invention, it has been adopted in animation, video games, and engineering. Simulating realistic cloth objects such as flags, curtains, and similar materials requires a physical model that captures how fabric stretches, folds, and responds to external forces such as wind. Among many continuous and discrete approaches to this problem, the mass-spring-damper model is one of the most widely used due to its balance between computational efficiency and realistic visuals.

In this model, a piece of cloth is represented as a 2D grid of particles positioned in 3D space. Neighboring particles are connected by springs modeled after Hooke's Law, producing forces that resist stretching, shearing, and bending. Dampers are attached to each spring to dissipate energy, preventing unrealistic, repeated oscillation. The motion of each particle is governed by a system of ordinary differential equations, where the acceleration of each particle is determined by computing the net force acting upon it, including connected springs and environmental forces, in this case gravity and wind.

The main challenge in mass-spring cloth simulation is the instability of numerical integrators. The coupled system of ODEs allows small changes in one particle to affect each of its neighbors; and so on, causing a ripple effect where errors compound throughout the system. Meanwhile, large spring constants impose tight restrictions on the integrator's time-step. The choice of numerical solver and its time-step directly determines the energy conservation accuracy and system divergence.

This project implements an interactive real-time 3D cloth simulation in Python using VPython for visualization. The cloth is configurable as a grid of particles connected by structural, shear, and bend springs, with real-time, user-adjustable parameters including gravity, particle mass, wind, integrator type, and time-step. Two numerical integrators are implemented: Euler's method and fourth-order Runge-Kutta method (RK4).

2 Related Work

Provot [3] helped establish the mass-spring approach for cloth simulation by using three types of springs for each particle in a grid: structural, shear, and bend springs. This is the same spring connection scheme used in this project. Provot also noted that achieving a realistic cloth stiffness requires large spring constants, which in turn require smaller time-steps to retain system stability. His solution was to implement spring deformation constraints after each step to limit how much a spring can stretch, eliminating the need for extremely large spring stiffness.

Baraff and Witkin [1] addressed the stability problem in cloth simulation in their 1998 paper. They demonstrated that methods like forward Euler need very small time-steps for stiff cloth systems and instead proposed a backward Euler integration. While their method is much more complex and involves solving a large linear system, it allows larger time-steps while keeping system stability. Although effective, this method is not implemented in this project due to its complexity and high computational cost.

3 Methodology

3.1 Physical Model

The cloth model is represented as a two-dimensional grid of $X \times Y$ particles with point-mass m in three-dimensional space. Each particle's state is described by six state variables: three for position $x \in \mathbb{R}^3$ and three for velocity $v \in \mathbb{R}^3$. The entire system state, comprised of the state variables of each particle, is stored in a one-dimensional array of length $6 \times X \times Y$.

The motion of each particle is described by Newton's second law:

$$f_{net} = m \cdot a, \quad a = \frac{dv}{dt}, \quad \frac{dx}{dt} = v$$

The net force on each particle is the sum of all forces acting upon it. Particles that are marked as pinned remain at fixed positions; their net force is set to 0 to keep them fixed in place.

$$f_{net} = \begin{cases} 0, & particle \in pinned \\ f_{spring} + f_{damping} + f_{gravity} + f_{wind}, & \text{else} \end{cases}$$

The **spring force** between two particles a and b is computed using Hooke's Law in three dimensions: $f_{spring} = k \cdot (\|x_b - x_a\| - l_{rest}) \cdot \frac{x_b - x_a}{\|x_b - x_a\|}$

The **spring damping force** between two particles a and b is computed by projecting the relative velocity of the two connected particles onto the spring's unit direction vector. $f_{damping} = -c \cdot \left((v_b - v_a) \cdot \frac{x_b - x_a}{\|x_b - x_a\|} \right) \cdot \frac{x_b - x_a}{\|x_b - x_a\|}$

The **force of gravity** on a particle is computed by multiplying the particle's mass m by gravity g; $f_{gravity} = m \cdot g, \quad$ where $g \in \mathbb{R}^3$.

The **force of wind** is computed by multiplying wind strength s by wind angle θ; $f_{wind} = s \times [\cos\theta, 0, \sin\theta]^T, \quad$ where $s, \theta \in \mathbb{R}$. A more physically accurate system would compute each face normal, project the wind onto it, and distribute those forces across the face's vertices.

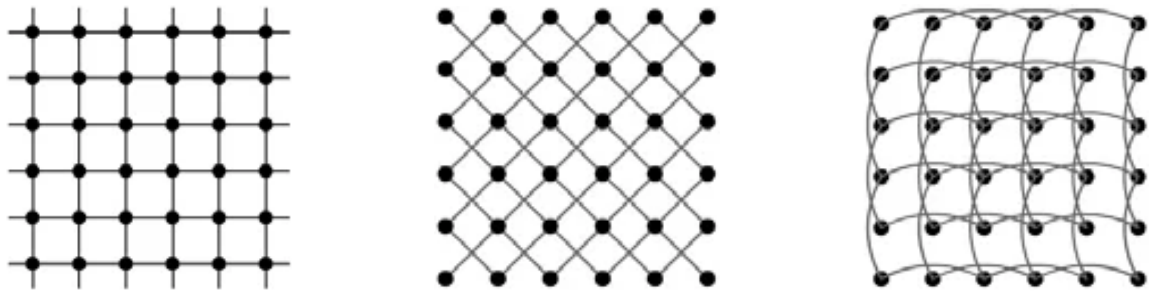
3.2 Spring Topology

The spring network connecting particles follows the topology introduced by Provot [3], using three types of springs that each resist a different type of deformation. All three types use the same Hooke's Law force equation described previously, differing only in which particles they connect, their rest lengths, and their spring constants.

Structural springs connect each particle to its immediate horizontal and vertical neighbors. These resist stretching and compression of the cloth along its grid axes. Their rest length is the initial grid spacing between particles d.

Shear springs connect each particle to its immediate diagonal neighboring particles. These resist the grid from deforming into a parallelogram shape. Their rest length is $d \cdot \sqrt{2}$.

Bend springs connect each particle to the particle two positions away horizontally and vertically, skipping one intermediate particle. These resist sharp folding of the cloth at individual particle joints. Their rest length is $2 \cdot d$.



(a) Structural spring (b) Shear spring (c) Bend spring

Figure 1: Spring types (source: [2])

The cloth system is expressed as a first-order ODE of the form $\frac{d(\text{state})}{dt} = f(\text{state})$, where the derivative function f computes the velocity and acceleration of every particle given the current state. The integrator advances the entire state vector as a single consistent update. This is required because the force on any given particle depends on the positions and velocities of its neighbors through shared springs, whose values must represent a consistent state.

Two integration methods are implemented. Euler's method computes the derivative at the current state and takes a single step forward:

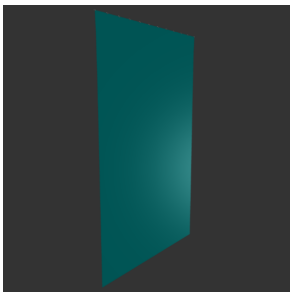
$$state_{n+1} = state_n + dt \cdot f(state_n)$$

Fourth-order Runge-Kutta (RK4) computes the derivative at four moments within each time-step to determine a more accurate estimate of the state after time t.

$$\begin{aligned}
k_1 &= f(state_n) \\
k_2 &= f(state_n + 0.5 \cdot dt \cdot k_1) \\
k_3 &= f(state_n + 0.5 \cdot dt \cdot k_2) \\
k_4 &= f(state_n + dt \cdot k_3) \\
state_{n+1} &= state_n + \frac{dt}{6}(k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned}$$

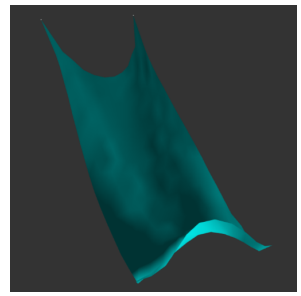
4 Results

The simulation produces plausible cloth behavior across a range of configurable parameters. Under gravity alone, the cloth drapes downward from its pinned top row. Adding wind causes the cloth to flap in the direction of the wind. Adjusting spring constants changes the cloth's behavior; higher k values produce a stiffer fabric that greatly resists stretching, while lower k values allow the cloth to deform much more readily under environmental forces.



(a) Gravity only; no wind. Top row particles pinned.

(b) Wind and gravity. Top row particles pinned.



(c) Low k value; cloth deforms. Top corner particles pinned.

Figure 2: Simulation results

4.1 Integrator Stability

To compare Euler with RK4, the simulation was run on a 10×10 cloth grid with $k = 200$, $m = 0.05$, and damping $c = 0.8$ for 1,000 time-steps at various values of dt. The system's frequency is $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{200}{0.05}} \approx 63.3$ rad/s. For a stiff oscillatory system such as this one, Euler's method requires a very small time-step to remain stable. In a coupled system with many springs, the effective stability limit becomes even more restrictive.

Table 1 shows the maximum particle velocity in the system after 1,000 steps for both integrators at each time-step. A stable simulation should show velocities near zero as damping dissipates energy. Divergence indicates the integrator is adding energy faster than damping can remove it. The step at which divergence occurs is noted where applicable.

Table 1 highlights several trends. At $dt = 0.001$, both integrators produce a stable

dt	Euler	RK4
0.001	0.2306 (stable)	0.1747 (stable)
0.005	diverged (step 975)	0.0021 (stable)
0.0075	diverged (step 532)	0.0001 (stable)
0.01	diverged (step 452)	0.0000 (stable)
0.015	diverged (step 304)	18.43 (unstable)
0.02	diverged (step 249)	diverged (step 146)

Table 1: Maximum particle velocity after 1,000 steps on a 10×10 grid ($k = 200$, $m = 0.05$, $c = 0.8$) with dt time-step size. Time-step of divergence is provided.

system; however, the Euler implementation has a 32% higher maximum velocity than the RK4 implementation, indicating that even at stable dt values, Euler's method introduces more numerical error than RK4. At $dt = 0.005$, the Euler system diverges while RK4 settles to near-zero velocity, demonstrating its stability at this time-step.

RK4 remains stable up to $dt = 0.01$, where it achieves a maximum velocity close to zero. At $dt = 0.015$, RK4's system produces a maximum velocity of 18.43, indicating that although it has not yet diverged, it is beginning to accumulate substantial error and is approaching instability. At $dt = 0.02$, both methods diverge.

The divergence data shows that both Euler's method and RK4 become unstable for sufficiently large dt values, but Euler's method remains stable over a smaller range of time-steps than RK4. At $dt = 0.02$, Euler's method diverges at step 249 while RK4 diverges sooner at step 146. This indicates that once the time-step is far beyond the stability range of both integrators, the behavior of the solution is dominated by numerical instability rather than accuracy, and different methods may diverge at different rates depending on their stability properties.

5 Discussion and Conclusions

This mass-spring-damper model demonstrated that for stiff oscillatory systems like cloth, integrator choice has a larger impact on simulation stability than it does for simpler, less coupled systems. Euler's method is effectively unusable at any time-step large enough for real-time performance, while RK4 provides stable results across a wider range of time-steps with the default simulation parameters.

The stability analysis confirms the theoretical expectation that the system's high natural frequency ($\omega \approx 63.3$ rad/s), combined with the coupling introduced by many springs, places strong demands on lower-order numerical solvers. Euler's lower-order approximation accumulates numerical error more quickly in the coupled system, contributing to rapid energy growth and divergence. RK4's fourth-order accuracy reduces numerical error enough for damping to counteract it, allowing stable behaviour for time-steps roughly up to $dt = 0.01$.

Other limitations include the absence of collision detection with rigid bodies and the constraint that very low particle masses ($m < 0.01$) still cause instability even with RK4, as the natural frequency $\omega = \sqrt{\frac{k}{m}}$ increases. Future work could address limitations through face-normal wind computation, sphere or plane collision response, and adaptive time-stepping that adjusts dt based on the current system stiffness.

All success conditions outlined in the project proposal were met. The simulation produces stable cloth dynamics under appropriate parameter choices, provides real-time interactive controls for physical parameters, implements both Euler and RK4 integration methods, and the analysis presented in this report quantifies the stability and performance trade-offs between them.

References

[1] D. Baraff and A. Witkin. Large steps in cloth simulation. In *Proceedings of SIGGRAPH*, pages 43–54, 1998.

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[4] D. Terzopoulos, J. Platt, A. Barr, and K. Fleischer. Elastically deformable models. In *Proceedings of SIGGRAPH*, pages 205–214, 1987.